Software Requirements Specification (SRS)

Student Attendance Management System

Project: Student Attendance Management System
Version: 1.0
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Revision History

Version	Date	Author	Change Summary	Approval
1.0	13-09-2026	Team 1	Initial SRS draft	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Student Attendance Management System. It defines the functional and non-functional requirements, system interfaces, security requirements, and verification criteria of the system. The document serves as a reference for developers, testers, faculty members, administrators, and other stakeholders involved in the development and evaluation of the system.

The main purpose of the Student Attendance Management System is to provide an efficient and centralized platform for recording, managing, monitoring, and reporting student attendance. The system reduces the need for manual attendance records and helps faculty and administrators maintain accurate attendance information.

1.2 Scope

The Student Attendance Management System is a web-based application designed to automate the process of student attendance management in an educational institution.

The system will allow authorized faculty members to:

- Log in securely to the system.
- View assigned classes and subjects.
- Record student attendance.
- Update attendance records when authorized.
- View attendance history.
- Calculate student attendance percentages.
- Generate attendance reports.
- Identify students with attendance below the required threshold.

Students will be able to:

- Log in to their accounts.
- View their attendance records.
- View attendance percentage for individual subjects.
- Monitor their attendance history.
- Receive attendance-related notifications where applicable.

Administrators will be able to:

- Manage student and faculty accounts.
- Manage subjects, classes, and courses.
- Assign faculty members to subjects/classes.
- View and generate overall attendance reports.
- Monitor attendance records and system activities.

1.3 Audience

This document is intended for the following stakeholders:

- **System Developers** – to understand and implement the required functionality.
- **Faculty Members** – to understand attendance recording and management features.
- **Students** – to understand how attendance information will be accessed and monitored.
- **System Administrators** – to manage users, subjects, classes, and attendance data.
- **QA/Test Engineers** – to verify that the system satisfies the specified requirements.
- **Project Evaluators/Faculty** – to assess whether the proposed system meets its functional, security, and quality requirements.

1.4 Definitions

Term	Definition
SRS	Software Requirements Specification
SAM	Student Attendance Management System
Admin	Authorized user responsible for managing the system
Faculty	Teacher/instructor responsible for recording and managing attendance
Student	User who can view their own attendance information
Attendance Record	Information indicating whether a student was present or absent for a particular class/session
Attendance Percentage	Percentage of attended classes out of the total conducted classes
Subject	Academic course or subject for which attendance is recorded
Session	A particular class period for which attendance is recorded
Authentication	Process of verifying a user's identity before granting system access
Authorization	Process of determining what actions an authenticated user is allowed to perform
UI	User Interface

API	Application Programming Interface
DB	Database
FR	Functional Requirement
NFR	Non-Functional Requirement

2. Overall Description

2.1 Product Perspective

The Student Attendance Management System is a web-based application designed to automate and centralize the process of recording and managing student attendance. The system will provide separate access based on user roles such as Administrator, Faculty, and Student. Faculty members will use the system to record and manage attendance, while students can view their attendance status and percentage. Administrators will manage users, subjects, classes, and other system information. The system will use a centralized database to store student details, faculty details, subjects, class information, attendance records, and attendance reports. The application can be developed using Python with Django and accessed through a web browser. The system will reduce manual paperwork, minimize errors in attendance calculations, and provide quick access to attendance information.

2.2 Major Product Functions

- **User Authentication** – Allows administrators, faculty, and students to securely log in.
- **Role-Based Access** – Provides different permissions based on the user's role.
- **Student Management** – Allows the administrator to add, update, view, and manage student details.
- **Faculty Management** – Allows the administrator to manage faculty accounts and information.
- **Subject Management** – Allows subjects to be created and assigned to appropriate classes.
- **Class Management** – Allows administrators to manage classes, sections, and student enrollment.
- **Attendance Marking** – Allows faculty members to mark students as present or absent.
- **Attendance Modification** – Allows authorized faculty members to correct attendance records.
- **Attendance History** – Allows users to view previously recorded attendance.

- **Attendance Percentage Calculation** – Automatically calculates attendance percentages for students.
- **Low Attendance Identification** – Identifies students whose attendance falls below the required percentage.
- **Attendance Reports** – Generates attendance reports for students, subjects, classes, or specified periods.
- **Student Attendance Dashboard** – Provides students with a summary of their attendance.
- **Notifications** – Provides attendance-related notifications or alerts to students.
- **Admin Dashboard** – Provides administrators with an overview of users and attendance information.
- **Data Management** – Provides controlled storage, updating, and retrieval of attendance-related information.

2.3 User Roles and Characteristics

The system will have three primary user roles.

Administrator

The Administrator has the highest level of access to the system. The administrator is responsible for managing users and system information.

The Administrator can:

- Create and manage student accounts.
- Create and manage faculty accounts.
- Manage subjects and classes.
- Assign faculty members to subjects/classes.
- View attendance information.
- Generate overall attendance reports.
- Manage system configurations.

Faculty

Faculty members are responsible for recording and managing attendance for the students assigned to them.

The Faculty can:

- Log in securely.
- View assigned subjects and classes.
- View enrolled students.
- Mark attendance.
- Modify attendance when authorized.
- View attendance history.
- Generate attendance reports.
- Identify students with low attendance.

Student

Students have limited access focused on their own attendance information.

The Student can:

- Log in securely.
- View their attendance records.
- View attendance percentage for each subject.
- View attendance history.
- Monitor low-attendance status.
- View attendance-related notifications.

2.4 Operating Environment

The Student Attendance Management System will operate in the following environment:

Software Environment:

- Operating System: Windows/Linux/macOS
- Backend: Python with Django
- Frontend: HTML, CSS, JavaScript
- Database: MySQL

2.5 Constraints

The following constraints apply to the system:

- Only authenticated users shall be allowed to access protected system functions.
- Users shall only be permitted to perform operations allowed by their assigned role.
- Attendance records shall be stored in a centralized database.
- Only authorized faculty members shall be allowed to record or modify attendance.
- Students shall only be able to view their own attendance information.
- The system requires a functioning network connection for accessing the web application.
- Attendance data must be protected from unauthorized modification or deletion.
- The system should maintain accurate attendance calculations based on the stored records.
- The system should be designed so that it can be extended with additional attendance methods in the future.

3. External Interface Requirements

3.1 User Interfaces

The Student Attendance Management System shall provide a simple and user-friendly web interface that can be accessed through a standard web browser.

The system shall provide separate interfaces based on the user's role.

Administrator Interface

The Administrator dashboard shall provide options to:

- Manage student accounts.
- Manage faculty accounts.
- Manage subjects and classes.
- Assign faculty to subjects/classes.
- View attendance information.
- Generate attendance reports.
- Monitor system activities.

Faculty Interface

The Faculty dashboard shall provide options to:

- View assigned subjects and classes.
- View the list of students.
- Mark attendance.
- Modify attendance when authorized.
- View attendance history.
- Generate attendance reports.
- View students with low attendance.

Student Interface

The Student dashboard shall provide options to:

- View personal attendance percentage.
- View subject-wise attendance.
- View attendance history.
- View attendance status.
- View attendance-related notifications.

The interface shall use clear navigation, readable text, appropriate forms, tables, buttons, and error messages. The system should also be responsive so that it can be accessed from desktops, laptops, tablets, and supported mobile devices.

3.2 Hardware Interfaces

The system shall support interaction with:

- Desktop computers and laptops used by administrators and faculty.
- Smartphones and tablets used by students, if mobile browser access is supported.
- Keyboard and mouse/touchscreen for user interaction.
- Server hardware or cloud infrastructure for hosting the application and database.

3.3 Software Interfaces

The Student Attendance Management System shall interact with the following software components:

- **Web Browser:** Google Chrome, Microsoft Edge, Mozilla Firefox, or equivalent.
- **Application Framework:** Python with Django.
- **Database Management System:** MySQL.
- **Operating System:** Windows, Linux, or macOS for development/deployment environments.
- **Authentication Module:** The system shall use an authentication mechanism to verify user credentials and control access according to user roles.

The application shall communicate with the database to store and retrieve user information, student details, subject information, and attendance records.

3.4 Communications

The system shall communicate between the user's web browser, application server, and database through appropriate network protocols.

The following communication requirements shall apply:

- The system shall use **HTTP/HTTPS** for communication between the client and web server.
- **HTTPS** should be used in the deployed system to protect user credentials and attendance information during transmission.
- Database communication shall be performed through secure database connections.
- The system shall provide appropriate error messages if communication with the server or database fails.
- The system shall validate data received from users before storing it in the database.
- Where email or notification services are used, communication shall take place through the appropriate secure API or service protocol.

4. System Features (Detailed)

This section defines the major functional requirements of the Student Attendance Management System (SAMS). Each requirement specifies what the system shall do, its priority, stakeholder, acceptance criteria, test case reference, and dependencies.

We will use the requirement IDs SAM-F-### for the functional requirements.

4.1 User Authentication and Authorization

Description: The system shall authenticate users and provide access according to their assigned roles.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Dependencies
SAM-F-001	The system shall allow registered users to log in using valid credentials.	Functional	High	Admin, Faculty, Student	Valid credentials successfully create a user session. Test: TC-AUTH-01	User database
SAM-F-002	The system shall reject login attempts containing invalid credentials.	Functional	High	Security	Invalid credentials shall not create a session and an appropriate error message shall be displayed. Test: TC-AUTH-02	Authentication module
SAM-F-003	The system shall provide role-based access for	Functional	High	Admin, Security	Each user shall only access functions	User roles

	Admin, Faculty, and Student users.				permitted for their role. Test: TC-AUTH-03	
SAM-F-004	The system shall allow authenticated users to securely log out of the system.	Functional	High	All Users	After logout, protected pages shall not be accessible without logging in again. Test: TC-AUTH-04	Authentication module

4.2 Student Management

Description: The system shall allow administrators to manage student information.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Dependencies
SAM-F-005	The system shall allow the administrator to add a new student with the required details.	Functional	High	Admin	A valid student record shall be stored successfully. Test: TC-STU-01	Student database
SAM-F-006	The system shall allow the administrator to update student information.	Functional	Medium	Admin	Updated information shall be reflected in the student's record. Test: TC-STU-02	Student database
SAM-F-007	The system shall allow the administrator to view and search	Functional	Medium	Admin, Faculty	The system shall display matching student records for a valid search.	Student database

	student records.				Test: TC-STU-03	

4.3 Faculty and Subject Management

Description: The system shall provide administrators with facilities to manage faculty, subjects, and assignments.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Dependencies
SAM-F-008	The system shall allow the administrator to create and manage faculty accounts.	Functional	High	Admin	A valid faculty account shall be created and made available for login. Test: TC-FAC-01	User database
SAM-F-009	The system shall allow the administrator to create and manage subjects.	Functional	High	Admin	A subject with valid details shall be stored successfully. Test: TC-SUB-01	Subject database
SAM-F-010	The system shall allow the administrator to assign faculty members to subjects and classes.	Functional	High	Admin	Assigned faculty shall be able to view the corresponding subject/class. Test: TC-FAC-02	Faculty and subject records
SAM-F-011	The system shall allow the administrator to manage student	Functional	High	Admin	Enrolled students shall appear in the appropriate faculty attendance	Student, class and subject data

	enrollment in classes or subjects.				list. Test: TC-ENR-01	

4.4 Attendance Management

Description: The system shall allow authorized faculty members to record and manage student attendance.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Dependencies
SAM-F-012	The system shall allow authorized faculty members to mark students as Present or Absent for a class session.	Functional	High	Faculty	Attendance shall be successfully recorded for all selected students. Test: TC-ATT-01	Faculty authorization, student list
SAM-F-013	The system shall prevent unauthorized users from marking attendance.	Functional	High	Security, Admin	Students and unauthorized users shall not be provided access to attendance marking. Test: TC-ATT-02	Role-based access
SAM-F-014	The system shall allow authorized faculty members to modify an attendance record when permitted.	Functional	High	Faculty, Admin	A permitted modification shall update the corresponding attendance record. Test: TC-ATT-03	Existing attendance record
SAM-F-015	The system shall prevent duplicate	Functional	High	Faculty, Admin	Attempting to create a duplicate	Attendance database

	attendance records for the same student, subject, date, and class session.				record shall be rejected or handled without creating duplicate data. Test: TC-ATT-04	
SAM-F-016	The system shall store the date, subject, class/session, student, and attendance status for each attendance record.	Functional	High	Admin, Faculty	A saved attendance record shall contain all required information. Test: TC-ATT-05	Attendance database

4.5 Attendance Viewing and Calculation

Description: The system shall provide students and faculty with appropriate attendance information.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Dependencies
SAM-F-017	The system shall allow students to view their own attendance records.	Functional	High	Student	A student shall only be able to view attendance associated with their own account. Test: TC-VIEW-01	Authentication, attendance database
SAM-F-018	The system shall calculate and display attendance percentage	Functional	High	Student, Faculty	The displayed percentage shall correspond to the stored attendance	Attendance database

	for each subject.				records. Test: TC-CALC-01	
SAM-F-019	The system shall allow faculty members to view attendance records for their assigned classes and subjects.	Functional	High	Faculty	Faculty shall see attendance records only for assigned classes/subjects. Test: TC-VIEW-02	Faculty assignment
SAM-F-020	The system shall identify students whose attendance is below the configured minimum attendance requirement.	Functional	High	Faculty, Admin	Students below the configured threshold shall be identified correctly. Test: TC-CALC-02	Attendance calculation

4.6 Attendance Reports and Notifications

Description: The system shall provide reporting and attendance-related alerts.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Dependencies
SAM-F-021	The system shall generate attendance reports based on student, subject, class, or date range.	Functional	High	Faculty, Admin	A report shall contain attendance information matching the selected criteria. Test: TC-REP-01	Attendance database
SAM-F-022	The system shall allow	Functional	Medium	Faculty, Admin	The system shall display	Attendance calculation

	authorized users to view attendance summaries and statistics.				correct attendance summaries for the selected class or subject. Test: TC-REP-02	
SAM-F-023	The system shall notify students when their attendance falls below the configured threshold.	Functional	Medium	Student, Faculty	A low-attendance student shall receive or see an appropriate notification. Test: TC-NOT-01	Notification module
SAM-F-024	The system shall maintain an attendance history for each student.	Functional	High	Student, Faculty, Admin	Previous attendance records shall remain available for authorized viewing. Test: TC-HIST-01	Attendance database

5. Non-Functional Requirements (Detailed)

This section defines the quality and performance requirements of the Student Attendance Management System.

We will use the requirement IDs SAM-NF-###.

5.1 Non-Functional Requirements

Req ID	Requirement	Category	Priority	Acceptance Criteria /

				Measurement
SAM-NF-001	The system shall display the requested attendance page within 3 seconds for at least 95% of requests under normal system load.	Performance	High	95th percentile response time ≤ 3 seconds. Test: TC-PERF-01
SAM-NF-002	The system shall provide at least 99% availability per month, excluding scheduled maintenance.	Reliability	High	Monthly uptime shall be ≥ 99%. Test: TC-REL-01
SAM-NF-003	The system shall ensure that users can access only the functions and data permitted by their assigned role.	Security	High	Unauthorized access attempts shall be denied. Test: TC-SEC-01
SAM-NF-004	The system shall provide a clear and easy-to-use interface so that a faculty member can mark attendance without requiring specialized technical knowledge.	Usability	Medium	Representative users shall successfully complete attendance marking without assistance. Test: TC-UX-01
SAM-NF-005	The system shall maintain the accuracy and consistency of attendance records when attendance is created, modified, or retrieved.	Data Integrity	High	Database records shall match the corresponding attendance transactions. Test: TC-DATA-01
SAM-NF-006	The system shall maintain activity logs for important administrative and attendance operations, including user, action, and timestamp where applicable.	Auditability	High	Required activities shall be recorded and retrievable by authorized administrators. Test: TC-AUDIT-01
SAM-NF-007	The system shall support concurrent access by at least 100 users without causing critical system errors under normal operating conditions.	Scalability	Medium	Load test with 100 concurrent users completes without critical failures. Test: TC-SCALE-01

5.2 Security

5.2.1 Security Objectives

The following security objectives define the main security goals of the Student Attendance Management System.

Security Objective 1 – Confidentiality

The system shall protect student, faculty, login, and attendance information from unauthorized access or disclosure. Users shall only be able to access information permitted by their assigned role.

Security Objective 2 – Integrity

The system shall protect attendance records from unauthorized or accidental modification. Only authorized users shall be permitted to create or modify attendance records, and important changes shall be recorded for auditing.

5.2.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test Case
SAM-SR-001	The system shall require authentication before allowing access to protected functions and student attendance information.	Security	High	An unauthenticated user attempting to access a protected page shall be redirected to the login page. Test: TC-SEC-01
SAM-SR-002	The system shall enforce role-based authorization for Administrator, Faculty, and Student users.	Security	High	A user shall be prevented from accessing functions outside their assigned role. Test: TC-SEC-02
SAM-SR-003	The system shall store user passwords using a secure one-way password hashing mechanism and shall not	Security	High	Database inspection shall confirm that passwords are not stored in plain text. Test: TC-SEC-03

	store passwords as plain text.			
SAM-SR-004	The system shall use HTTPS for protecting credentials and other sensitive information during communication in the deployed environment.	Security	High	Network inspection shall confirm that protected application traffic uses HTTPS. Test: TC-SEC-04
SAM-SR-005	The system shall record important attendance and administrative operations with the responsible user and timestamp where applicable.	Security/Audit	High	Audit records shall identify the user and time associated with the selected operation. Test: TC-SEC-05

6. Quality Attributes & Acceptance Tests

6.1 Quality Attributes

The Student Attendance Management System shall satisfy the following quality attributes:

Quality Attribute	Requirement
Performance	The system should respond to normal attendance-related requests within 3 seconds for at least 95% of requests under normal load.
Reliability	The system should maintain at least 99% availability per month, excluding scheduled maintenance.
Security	Only authenticated and authorized users should access attendance and user information according to their assigned roles.
Usability	The interface should be simple and understandable so that faculty can mark attendance without specialized technical knowledge.
Data Integrity	Attendance records must remain accurate and consistent when they are created, modified, or retrieved.
Auditability	Important administrative and attendance operations should be logged with the responsible user and timestamp.
Scalability	The system should support at least 100 concurrent users under normal operating conditions.

Maintainability	The software should use modular components so that attendance, user management, reporting, and authentication can be maintained independently.

6.2 Acceptance Tests

Test Case ID	Requirement Tested	Acceptance Criteria	Expected Result
TC-AUTH-01	SAM-F-001	Enter valid username and password	User is successfully logged in
TC-AUTH-02	SAM-F-002	Enter invalid credentials	Login is rejected and an error is displayed
TC-AUTH-03	SAM-F-003	Access functions using different roles	Only permitted functions are accessible
TC-ATT-01	SAM-F-012	Faculty marks students Present/Absent	Attendance is stored correctly
TC-ATT-02	SAM-F-013	Student attempts to mark attendance	Access is denied
TC-ATT-04	SAM-F-015	Submit attendance twice for same session	Duplicate record is prevented
TC-VIEW-01	SAM-F-017	Student opens attendance page	Only the student's own attendance is displayed
TC-CALC-01	SAM-F-018	Compare displayed percentage with attendance records	Percentage is calculated correctly
TC-CALC-02	SAM-F-020	Use records below the configured threshold	Low-attendance students are identified
TC-REP-01	SAM-F-021	Generate report using selected criteria	Report contains the correct records
TC-SEC-03	SAM-SR-003	Inspect stored passwords	Passwords are not stored in plaintext
TC-SEC-04	SAM-SR-004	Access deployed system	Communication uses HTTPS

6.3 Acceptance Criteria

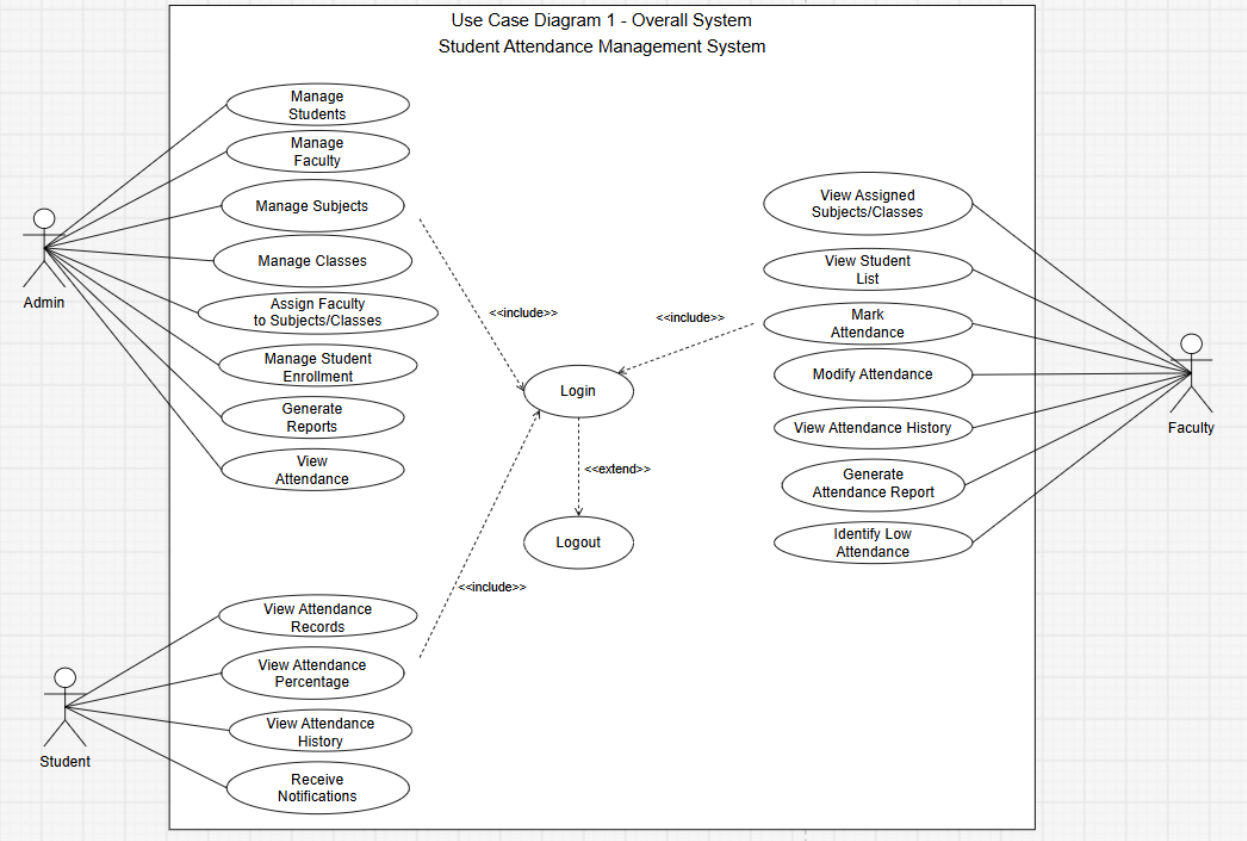
The system shall be considered acceptable when:

1. All 24 functional requirements are successfully implemented and tested.
2. All 7 non-functional requirements meet their specified acceptance criteria.
3. The 2 security objectives are satisfied.
4. All 5 security requirements pass their corresponding security tests.
5. Role-based access correctly separates Admin, Faculty, and Student privileges.

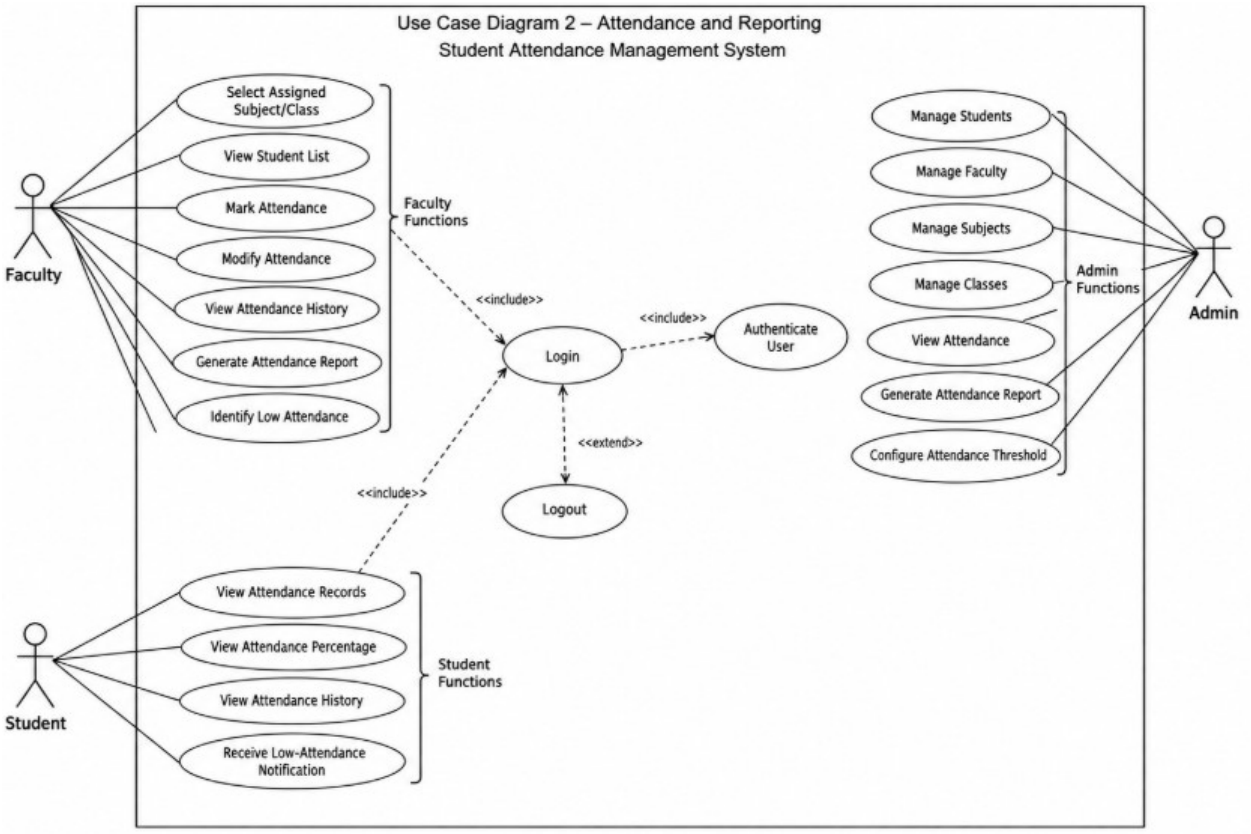
6. Attendance records and calculated percentages are accurate.
7. Required reports can be generated successfully.
8. No critical defects remain in the final release.

7. UML Use-Case Diagrams

7.1 Use-Case Diagram 1 – Overall System



7.2 Use-Case Diagram 2 – Attendance and Reporting



8. Requirements Traceability Matrix (RTM)

8.1 Functional Requirements

Req ID	Short Requirement	Section / Design Specification	Module	Test Case(s)	Status (N/P/A)	Comments
SAM-F-001	User login	4.1 Authentication	Authentication	TC-AUTH-01	N	Valid credentials required
SAM-F-002	Reject invalid login	4.1 Authentication	Authentication	TC-AUTH-02	N	Invalid credentials rejected

SAM-F-003	Role-based access	4.1 Authorization	Authorization	TC-AUTH-03	N	Admin/Faculty/Student
SAM-F-004	User logout	4.1 Authentication	Authentication	TC-AUTH-04	N	Session terminated
SAM-F-005	Add student	4.2 Student Management	Student Management	TC-STU-01	N	Admin only
SAM-F-006	Update student	4.2 Student Management	Student Management	TC-STU-02	N	Admin only
SAM-F-007	Search student	4.2 Student Management	Student Management	TC-STU-03	N	Authorized users
SAM-F-008	Manage faculty	4.3 Faculty Management	Faculty Management	TC-FAC-01	N	Admin only
SAM-F-009	Manage subjects	4.3 Subject Management	Subject Management	TC-SUB-01	N	Admin only
SAM-F-010	Assign faculty	4.3 Faculty Management	Assignment Management	TC-FAC-02	N	Faculty assigned to classes
SAM-F-011	Manage enrollment	4.3 Enrollment	Enrollment Management	TC-ENR-01	N	Student-class association
SAM-F-012	Mark attendance	4.4 Attendance Management	Attendance	TC-ATT-01	N	Faculty only
SAM-F-01	Prevent unauthorized	4.4 Attendance Managemen	Authorization	TC-ATT-02	N	Unauthorized access denied

3	attendance marking	t				
SAM-F-014	Modify attendance	4.4 Attendance Management	Attendance	TC-ATT-03	N	Authorized modification
SAM-F-015	Prevent duplicates	4.4 Attendance Management	Attendance	TC-ATT-04	N	One record per session
SAM-F-016	Store attendance details	4.4 Attendance Management	Attendance Database	TC-ATT-05	N	Date, student, subject, status
SAM-F-017	View own attendance	4.5 Attendance Viewing	Student Dashboard	TC-VIEW-01	N	Student sees own data
SAM-F-018	Calculate percentage	4.5 Attendance Calculation	Attendance Calculation	TC-CALC-01	N	Subject-wise percentage
SAM-F-019	Faculty attendance view	4.5 Attendance Viewing	Faculty Dashboard	TC-VIEW-02	N	Assigned classes only
SAM-F-020	Identify low attendance	4.5 Attendance Calculation	Attendance Monitoring	TC-CALC-02	N	Uses configured threshold
SAM-F-021	Generate reports	4.6 Reports	Reporting	TC-REP-01	N	Student/subject/ class/date
SAM-F-022	View summaries	4.6 Reports	Reporting	TC-REP-02	N	Attendance statistics
SAM-F-023	Low-attendance notification	4.6 Notifications	Notification	TC-NOT-01	N	Student notification
SAM	Maintain	4.6	Attendance	TC-	N	Historical records

-F-024	attendance history	Attendance History	History	HIST-01		retained

8.2 Non-Functional Requirements

Req ID	Short Requirement	Section / Design Specification	Module	Test Case(s)	Status	Comments
SAM-NF-001	Response within 3 seconds	5.1 Performance	Application	TC-PERF-01	N	≥95% normal requests
SAM-NF-002	99% availability	5.1 Reliability	System Infrastructure	TC-REL-01	N	Excluding scheduled maintenance
SAM-NF-003	Secure role-based access	5.1 Security	Authorization	TC-SEC-01	N	Prevent unauthorized access
SAM-NF-004	Easy-to-use interface	5.1 Usability	User Interface	TC-UX-01	N	Simple attendance workflow
SAM-NF-005	Maintain data integrity	5.1 Data Integrity	Database	TC-DATA-01	N	Accurate attendance records
SAM-NF-006	Maintain activity logs	5.1 Auditability	Audit Logging	TC-AUDIT-01	N	Important actions recorded
SAM-NF-007	Support 100 concurrent users	5.1 Scalability	System Infrastructure	TC-SCALE-01	N	Normal operating conditions

8.3 Security Requirements

Req ID	Short Requirement	Section / Design Specification	Module	Test Case(s)	Status	Comments
SAM-SR-001	Authentication required	5.2 Security	Authentication	TC-SEC-01	Planned	Protected functions require login
SAM-	Role-based authorization	5.2 Security	Authorization	TC-SEC-0	N	Admin/Faculty/ Student permissions

SR-002				2		
SAM-SR-003	Secure password hashing	5.2 Security	Authentication	TC-SEC-03	N	No plaintext passwords
SAM-SR-004	HTTPS communication	5.2 Security	Communication Security	TC-SEC-04	N	Required for deployment
SAM-SR-005	Security/ activity logging	5.2 Security	Audit Logging	TC-SEC-05	N	User and timestamp recorded